

Title: Body composition assessment and sarcopenia in patients with biliary tract cancer:

A systematic review and meta-analysis (protocol)

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1.Introduction

Biliary tract cancers (BTC) include intrahepatic (ICC) and extrahepatic cholangiocarcinoma (ECC), perihilar cholangiocarcinoma (PHC), gallbladder carcinoma (GC), and ampullary carcinoma [1–3]. BTC is common in eastern Asia and an aggressive malignancy with a poor prognosis, with less than half of patients presenting with unresectable disease and a 5-year overall survival (OS) rate of only 5–15% [4]. Its main treatment modality is surgical resection, and the elderly population undergoing surgical resection is steadily increasing.

With the aging of the population, body composition to assess sarcopenia has been attracting attention. Sarcopenia, described as an age-related decrease in skeletal muscle mass, has recently been a poor prognostic factor in patients after surgery for various malignancies [5–8]. Assessment of body composition for sarcopenia includes computed tomography (CT) scans, which are routinely used in clinical practice, and dual-energy X-ray absorptiometry (DXA) and bioelectrical impedance (BIA), which are less commonly used in cancer patients. However, there were no systematic review of the impact of assessment of body composition in patients with biliary tract cancer.

Therefore, the purpose of the study will be to assess impact of assessment of body composition in patients with biliary tract cancer.

2. Research question

P: Patients with BTC including ICC, ECC, PHC, GC, and ampullary carcinoma.

E: diagnosed with sarcopenia by body composition

C: diagnosed without sarcopenia by body composition

O: Overall survival (OS), recurrence-free survival (RFS), postoperative complications categorized according to the Clavien-Dindo (CD) $\geq$ III [10].

3. Method

3.1 The Preferred reporting items for systematic review and meta-analysis 2020 (PRISMA-2020) were followed for preparing this protocol [11]. This protocol will be published in protocols.io (<https://www.protocols.io/>).

3.2 Inclusion criteria of the articles for the review

3.2.1 Type of studies

We will include studies that assess of body composition in patients with biliary tract cancer. We will not apply language or country restrictions. We will include all papers including published, unpublished articles, abstract of conference and letter. We will exclude review articles, and case reports. We will not exclude studies based on the observation period or publication year.

3.2.2 Study participants

Inclusion criteria will be patients over the age of 18 years with biliary tract cancer.

Exclusion criteria will be mixed populations in which the outcome of patients with cancer at another site (pancreatic cancer and/or hepatic cancer) cannot be separated from those of patients with biliary tract cancer.

3.2.3 Exposure

Exposure will be diagnosed with sarcopenia by body composition assessed by any method of reporting either the volume or characteristics of muscle and/or adipose

compartments within the body. Body composition will be measured by method of assessment (computed tomography [CT], dual-energy X-ray absorptiometry [TX] bioelectrical impedance [BIA]) definition and cut-off points defining sarcopenia.

3.2.4 Control

Control will be diagnosed without sarcopenia by body composition.

3.3 Type of outcomes

3.3.1 Primary outcomes

1. Overall survival (OS)
2. Recurrence-free survival (RFS)
3. Post-operative major complications (CD $\geq$ III).

3.4 Search method

3.4.1 Electronic search

We will search the following databases: MEDLINE (PubMed); the Cochrane Central

Register of Controlled Trials (Cochrane Library); and EMBASE (Dialog) (Appendix 1).

3.4.2 Other resources

We will also search the following databases for ongoing or unpublished trials: the World Health Organization International Clinical Trials Platform Search Portal (ICTRP); and ClinicalTrials.gov (Appendix 2).

We will check the reference lists of studies, including international guidelines [12–16] as well as the reference lists of eligible studies and articles citing eligible studies. We will ask the authors of original studies for unpublished or additional data.

3.5 Data collection and analysis

3.5.1 Selection of the studies

Two independent reviewers (JW and RM) will screen titles and abstracts, followed by the assessment of the eligibility based on the full texts. We will contact original authors if relevant data is missing. Disagreements between the two reviewers will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (KK).

3.5.2 Data extraction and management

Two reviewers (JW and RM) will perform independent data extraction of the included studies using standardized data collection form. We will use a pre-checked form using 10 randomly selected studies. The form will include the information on study design, study population, exposures and outcomes. Any disagreements will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (KK).

3.6 Assessment of risk of bias in included studies

Two reviewers (JW and RM) will evaluate the risk of bias independently using the Newcastle-Ottawa Quality Rating Scale (NOS) [17]. Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer (KK) will be acting as an arbiter, if necessary.

3.7 Measures of treatment effects

We will pool the relative risk ratios and the 95% confidence intervals (CIs) for the

following binary variables: Post-operative major complications.

We will pool the hazard ratios and the 95% CIs for the following continuous variables:

OS and RFS.

3.8 Handling of missing data

We will ask not-presented data to the original authors.

3.8.1 Missing outcomes

We will perform the intention-to-treat (ITT) analysis for all dichotomous data as much as possible. For continuous data, we will not impute missing data based on the recommendation by Cochrane handbook [18]. We will perform meta-analysis about the available data in the original study.

3.8.2 Missing statistics

When original studies only report standard error or p-value, we will calculate the standard deviation based on the method by Altman [19]. If we don't know these values when we

contact the authors, standard deviation will be calculated by confidence interval and t-value based on the method by Cochrane handbook [18], or validated method [20]. Validity of these methods will be analyzed by sensitivity analysis.

3.10 Assessment of heterogeneity

We will evaluate the statistical heterogeneity by visual inspection of the forest plots and calculating the I^2 statistic (I^2 values of 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity) [18]. When there is substantial heterogeneity ($I^2 > 50\%$), we will assess the reason of the heterogeneity. Cochrane χ^2 test (Q-test) will be performed for I^2 statistic, and P value less than 0.10 will be defined as statistically significant.

3.11 Assessment of reporting bias

We will search the clinical trial registry system (ClinicalTrials.gov and ICTRP) and will perform extensive literature search for unpublished trials. To assess outcome reporting

bias, we will compare the outcomes defined in trial protocols with the outcomes reported in the publications. We will assess the potential publication bias by visual inspection of the funnel plot. We will not conduct the test when we find less than 10 trials or trials which have similar sample size. We will also assess the potential publication bias by visual inspection of the funnel plot.

3.12 Meta-analysis

Meta-analysis will be performed using Review Manager software (RevMan 5.4.2). We will use a random-effects model.

3.13 Subgroup analysis

To elucidate the influence of effect modifiers on results, we will evaluate the subgroup analyses of the primary outcomes on the following factors when sufficient data are available.

1. Cancer types (ICC, ECC, PHC, GC, or ampullary carcinoma.)
2. Body composition measures (CT, DXA, or BIA)

3.14 Sensitivity analysis

We will undertake the following sensitivity analyses for the primary outcomes to assess whether the results of the review are robust to the decisions made during the review process.

1. Exclusion of studies using imputed statistics.
2. Only the participants who complete the study with complete data.

4. Summary of findings table

Two reviewers (JW and RM) will evaluate the certainty of evidence based on the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach [21].

Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer (KK) will be acting as an arbiter, if necessary.

Summary of findings table will be made for the following outcome based on the Cochrane handbook [18]: OS, RFS, and major complications.

5. Conflict of Interest

The authors declare no conflicts of interests.

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Appendix 1: Search strategy

MEDLINE (PubMed) search strategy

#1. "Biliary Tract Neoplasms"[Mesh] OR "Bile Duct Neoplasms"[Mesh] OR

"Gallbladder Neoplasms"[Mesh] OR "Carcinoma, Papillary"[Mesh]

#2. ((Bil*[tiab] OR cholangio*[tiab]) AND (carcinom*[tiab] OR cancer*[tiab] OR
neoplasm*[tiab] OR malign*[tiab] OR tumo*[tiab]))

#3. #1 OR #2

#4. "Sarcopenia"[Mesh] OR "Muscle Weakness"[Mesh] OR "Muscle Weakness"[Mesh]

#5. "Sarcopenia"[tiab] OR "Muscular Atrophy"[tiab] OR "Muscle Weakness"[tiab]

#6. #4 OR #5

#7. "Body Composition"[Mesh]

#8. "Body Composition"[tiab]

#9. #7 OR #8

#10. #6 OR #9

#11. #3 AND #10

CENTRAL (Cochrane Library) search strategy

#1. MeSH descriptor: [Biliary Tract Neoplasms] explode all trees

#2. MeSH descriptor: [Bile Duct Neoplasms] explode all trees

#3. MeSH descriptor: [Gallbladder Neoplasms] explode all trees

#4. MeSH descriptor: [Carcinoma, Papillary] explode all trees

#5. ((Bil* OR cholangio*) AND (carcinom* OR cancer* OR neoplasm* OR malign* OR tumo*)):ti,ab,kw (Word variations have been searched)

#6. #1 OR #2 OR #3 OR #4 OR #5

#7. MeSH descriptor: [Sarcopenia] explode all trees

#8. MeSH descriptor: [Muscle Weakness] explode all trees

#9. MeSH descriptor: [Muscular Atrophy] explode all trees

#10. ("Sarcopenia" OR "Muscular Atrophy" OR "Muscle Weakness"):ti,ab,kw (Word variations have been searched)

#11. #7 OR #8 OR #9 OR #10

#12. MeSH descriptor: [Body Composition] explode all trees

#13. "Body Composition":ti,ab,kw (Word variations have been searched)

#14. #12 OR #13

#15. #11 OR #14

#16. #6 AND #15

EMBASE (Dialog) search strategy

S1 (EMB.EXACT.EXPLODE("biliary tract tumor"))

S2 (EMB.EXACT.EXPLODE("bile duct tumor"))

S3 (EMB.EXACT.EXPLODE("gallbladder tumor"))

S4 (EMB.EXACT.EXPLODE("papillary carcinoma"))

S5 ab((Biliary OR cholangio) AND ab(carcinoma OR cancer OR neoplasm OR
malignancy OR tumor)) OR ti((Biliary OR cholangio) AND ti(carcinoma OR cancer
OR neoplasm OR malignancy OR tumor))

S6 S1 OR S2 OR S3 OR S4 OR S5

S7 EMB.EXACT.EXPLODE("sarcopenia")

S8 (EMB.EXACT.EXPLODE("muscle weakness"))

S9 (EMB.EXACT.EXPLODE("muscle atrophy"))

S10 ab("Sarcopenia" OR "Muscular Atrophy" OR "Muscle Weakness") OR

ti("Sarcopenia" OR "Muscular Atrophy" OR "Muscle Weakness")

S11 S7 OR S8 OR S9 OR S10

S12 (EMB.EXACT.EXPLODE("body composition"))

S13 ab("body composition") OR ti("body composition")

S14 S12 OR S13

S15 S11 OR S14

S16 S6 AND S15

Appendix 2: Search strategy for clinical trial registry

ICTRP search strategy

"Biliary Tract Neoplasms" AND ("Sarcopenia" OR "Body Composition")

ClinicalTrials.gov search strategy

"Biliary Tract Neoplasms" AND ("Sarcopenia" OR "Body Composition")